

PRECISION CARE CHALLENGE 2026 · PHASE 1, IDEA SUBMISSION

ClaimCast

Know what your hospital bill will actually cost you. **Before** you are admitted, not **after** you are discharged.

TRACK: HOSPITALITY · Holistic Optimization System for Policy-Integrated Admission & Treatment Intelligence

Team Rocket

(aniket22073)

Indraprastha Institute of Information Technology, Delhi (IIIT-Delhi)

Explain The Problem You Are Attempting to Solve

India's most expensive healthcare decision is made in ninety seconds, at an admission counter, by someone who has never read their policy.

IRDAI's chairman has said claim payouts often fall short of what policyholders expect.

AJAY SETH, CHAIRMAN, IRDAI · NOV 2025

The regulator has named the gap. Policy bots explain the wording; none of them price the decision in front of you.

Proportionate deduction · IRDAI/HLT/REG/CIR/151/06/2020 · maximal case; scope varies by policy wording

THE ROOM-RENT TRAP

50% of the bill can be refused

Your policy caps the room at **₹5,000** a day. The hospital gave you one at **₹10,000**. Surgeon, OT, anesthesia, nursing: **every charge your policy ties to room category** is scaled down by that same ratio.

₹39,085

mean out-of-pocket per **cancer hospitalization**
12,148-patient multi-centre Indian study

₹3.31 L

out-of-pocket per **cancer patient, per year**
diagnostics 36% · medicines 45%

Cost shock drives distress financing, delayed care and outright **treatment abandonment**. It decides whether treatment finishes.

2

Problems solved

BASE BILL ESTIMATION

Forecasts your itemized bill from public tariff data, before you're admitted.

INSURANCE ANALYSIS

Computes exactly what your policy refuses, line by line, cited to the clause.

Briefly Explain Your Idea/Solution

Insurers compute what you are owed **after discharge**. ClaimCast computes it **before admission**.

Same arithmetic. Different timestamp. That is the whole idea.

1 · ADMISSION

Which hospital

gap between in-network and out-of-network payout

₹1.2 L

at stake

2 · ADMISSION

Which room class

private room vs. semi-private, after proportionate deduction

₹78,400

at stake

3 · INVESTIGATION

Cashless or reimburse

cash you must float upfront if not cashless

₹2 L

at stake

4 · PROCEDURE

Implant / consumable

cost of implants excluded under List I

₹45,000

at stake

5 · RECOVERY

24-hour admission rule

lost if care is billed day-care, not in-patient

₹22,000

at stake

THE CARE JOURNEY THE PROBLEM STATEMENT NAMES

we price the decision at every stage, and each fork closes permanently the moment it passes

CONSTRAINT OPTIMIZATION

The model re-runs your claim across every legal alternative and ranks them by what you keep, filtering out anything that breaks a constraint — Room-rent cap **₹10,000** , waiting period **1 yr**.

IS THIS UNIQUE, OR AN IMPROVISATION?

Unique. Existing tools **retrieve** policy text. ClaimCast **simulates the claim and optimizes the decision**. Retrieval → Simulation.

YOU WALK IN KNOWING WHAT YOU WILL OWE. Same admission, ₹78,400 less out of pocket, because the caregiver was told before the decision closed.

Explain Your Solution

We estimate what you'll be billed. We compute exactly what won't be paid.

AI where uncertain.
Arithmetic where known.
The LLM never decides the money.

TWO HALVES

A · FORECAST

Predicts the itemised bill from published tariff data. Uncertain, so it returns a range.

B · ADJUDICATION

Computes what the policy refuses, line by line. Exact, with every ₹ cited to its clause.
COUNTERFACTUAL Re-runs the claim for every legal alternative and ranks them by what you keep.

AGAINST THE CONVENTIONAL ALTERNATIVES

	Hospital / TPA desk	Policy bots · Ditto/PB	ClaimCast
Speaks	after admission	any time, no context	at each decision
Returns	a verbal estimate	a text summary	itemized ₹ + clause
Acts for	the hospital	no one	you

PROS

- Deterministic and auditable
- Runs entirely on public data
- Rule-packs: private, PM-JAY, ESI

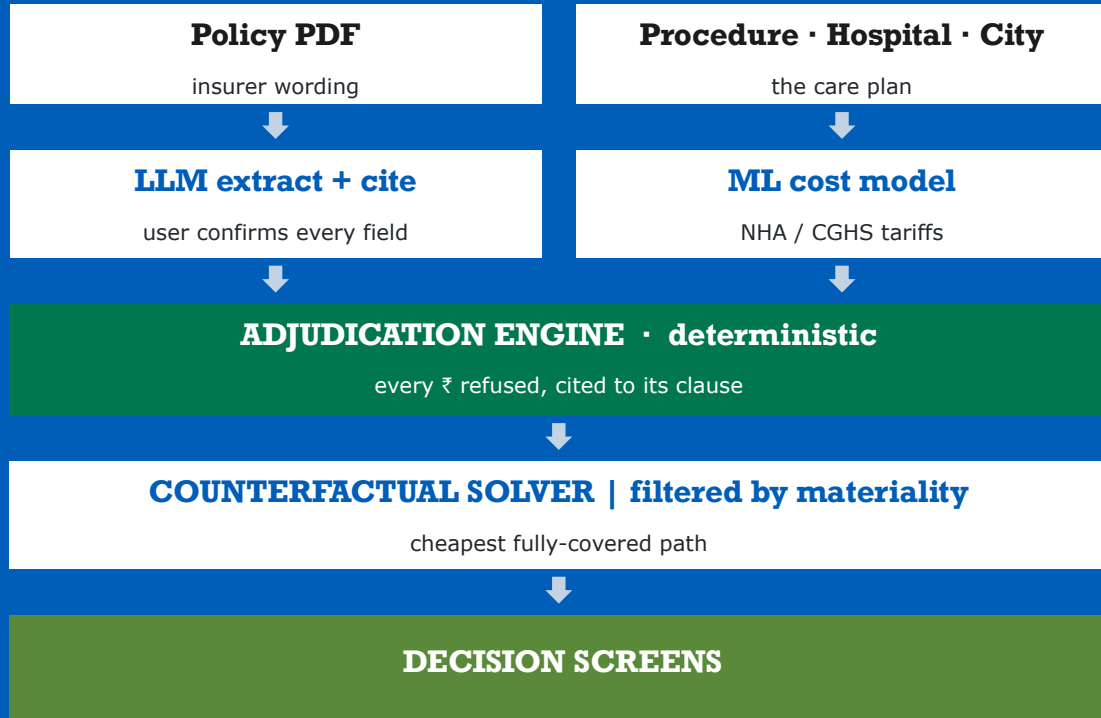
CONS

- Forecast bounded by published tariffs
- Needs the policy document up front
- An estimate, never a guarantee

ClaimCast				DECISION 2 OF 5 · ROOM CLASS · STAGE: ADMISSION		PROTOTYPE SCREEN	
○	Private room	₹10,000 / day	your share	₹1,26,900	← recommended by ClaimCast		
●	Semi-private	₹5,000 / day	your share	₹48,500			
✓ You keep ₹78,400 Room Rent Sub-limit, Clause 3.2 · proportionate deduction on the associated charges							

Architecture & AI Pipeline

ARCHITECTURE



REFERENCES

1. IRDAI, Circular IRDAI/HLT/REG/CIR/151/06/2020, room-rent proportionate deduction.
2. Financial toxicity of cancer treatment in India, Front. Public Health, 2023 (n = 12,148).
3. Catastrophic expenditure & treatment attrition, Lancet Reg. Health SE Asia, 2022;6:100058.
4. IRDAI Master Circular on Health Insurance Business, 2024.

ML cost model

XGBoost quantile regression — predicts a cost range, not a point estimate
Trained on: NHA/PM-JAY rates ($\approx 1,900$ procedures) · CGHS city-wise tariffs · past forecast-vs-actual outcomes (self-learning loop)

LLM extract + cite

A frontier LLM, swapped behind an interface — RAG over the uploaded policy PDF
Extracts: sum insured, sub-limits, co-pay %, exclusions, waiting periods — each field cited to its source clause
Extraction only, never adjudication — a misread field fails safe, flagged for user confirmation, never silently changes what's refused

MEMORY · SELF-LEARNING LOOP

Logs (procedure, hospital, city, forecast, actual billed amount) per case. Retrains the cost model on the combined historical + incoming pool — the range narrows as volume grows.

DATA WE SOURCE OURSELVES · ALL PUBLIC

- NHA / PM-JAY Health Benefit Packages ($\approx 1,900$ procedure rates)
- CGHS city-wise rate lists · NHA empanelled-hospital registry
- IRDAI standard wordings (Arogya Sanjeevani) · IRDAI List I non-payables

THANKYOU

ClaimCast

The decision, before the bill.